CSE 111 Fall 2026: Project Checkpoint 1

UCM Course Catalog & Registration Manager

Alyssa Phouangpanya, Aman Kalkat

Synopsis

This project targets university course registration, enrollment, and academic planning at UC Merced. The database will use data from UC Merced's publicly available course catalog, paired with synthetic student, faculty, and enrollment data. The application addresses any difficulty with coordinating prerequisites, section capacity, and student eligibility for courses through a unified application. The primary users of this application would be students checking eligibility and enrolling in classes, academic advisors verifying prerequisite completion, and registrars managing capacity, instructor assignments, and enrollments.

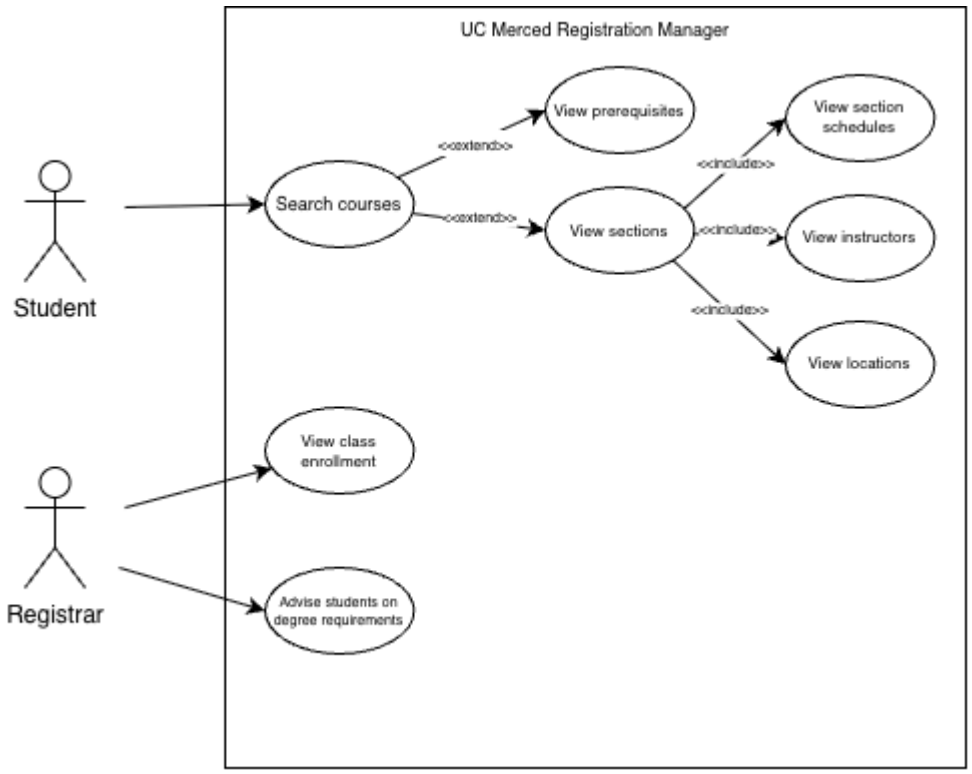
Authors & Contribution: Alyssa Phouangpanya: 50%, Aman Kalkat: 50%

UML Use Case Diagram

Students and academic advisors can utilize the system. Students can view courses and relevant course information such as prerequisites and available sections to filter classes that match their needs and that they are eligible for. Academic advisors can utilize the above data to assist students in finding courses that fulfill degree requirements and manage enrollment of students in course sections.

Authors & Contribution: Alyssa Phouangpanya: 50%, Aman Kalkat: 50%

Use Case Diagram:

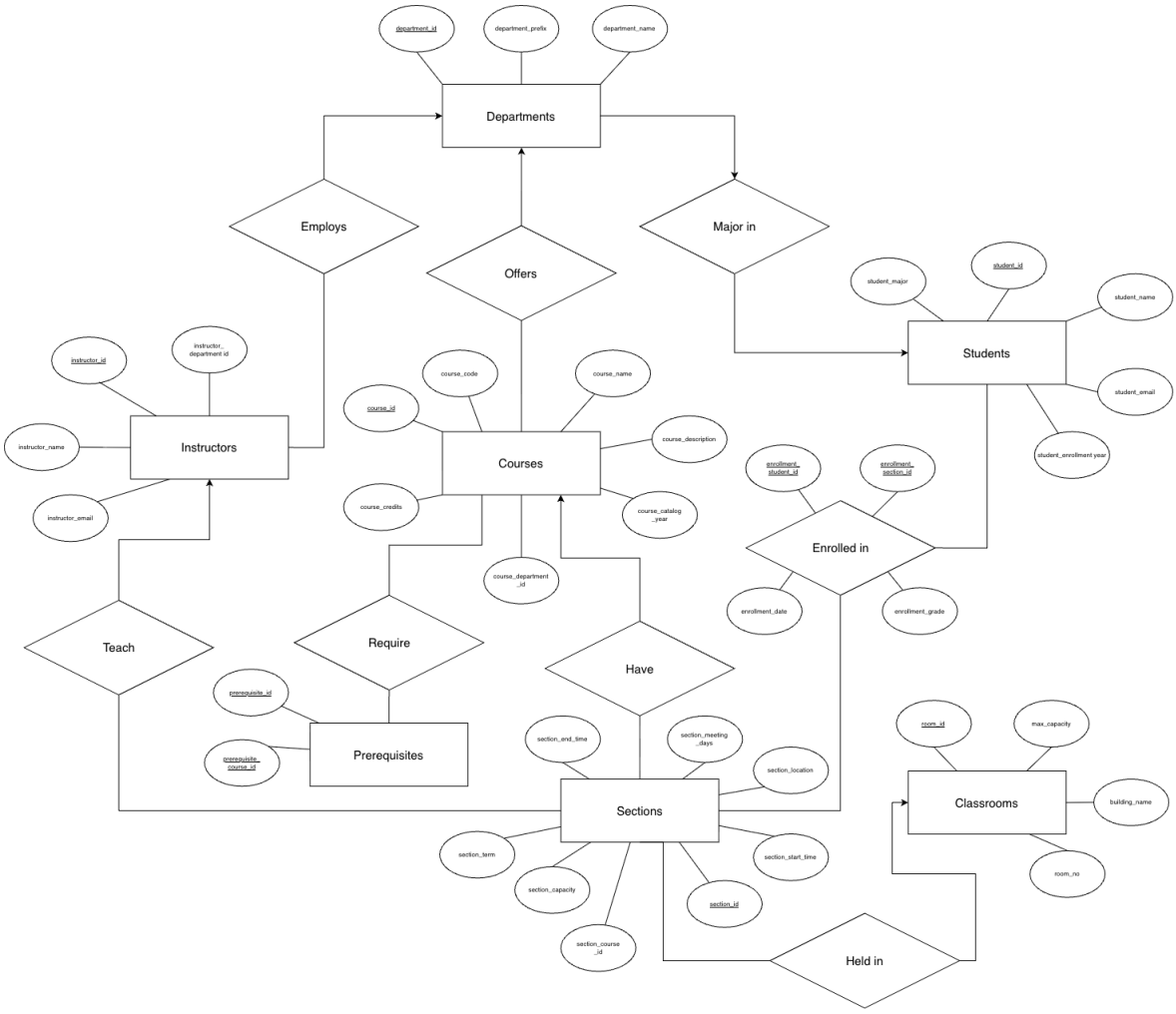


Authors & Contribution: Alyssa Phouangpanya: 80%, Aman Kalkat: 20%

ER Diagram

Our ER diagam consists of 7 entities connected by 8 relationships. The entities are Departments, Instructors, Students, Courses, Prerequisites, Sections, and Classrooms. Every entity has a primary key and no weak entities are present. The 2 many to many relationships are “Enrolled In” between Students and Sections as well as “Require” between Courses and their Prerequisites. The remaining relationships are: “Employs” connecting Instructors to their Departments, “major in” for Students to departments, “offers” for departments and their courses,” teaches” for instructors and sections, “have” that shows sections courses offer, and “held in” for what classroom those sections are in.

ERD-1:



Authors & Contribution: Alyssa Phouangpanya: 50%, Aman Kalkat: 50%

Relational Schema

Relational schema is made of up 8 realtions: Students, Instructos, Departments, Courses, Prerequisites, Enrollments, Sections, and Classrooms. Departments is one to many to Instructors, Courses, and Students (as their major). Courses is one to many to Sections, while Instructors and Rooms are each on to many to sections. Students and Sections are many to many through the Enrollments entity and Courses is many to many with Prerequisites.

Authors & Contribution: Alyssa Phouangpanya: 50%, Aman Kalkat: 50%

Students

CREATE TABLE Students (

student_id INTEGER PRIMARY KEY,

student_firstname VARCHAR(20) NOT NULL,

student_lastname VARCHAR(20) NOT NULL,

student_email VARCHAR(40) UNIQUE NOT NULL,

Student_enrollment_year DECIMAL(4,0)

Student _*major* _deparment_id INTEGER

);

Authors & Contribution: Aman Kalkat 100%

Instructors

CREATE TABLE Instructors (

instructor_id INTEGER PRIMARY KEY

Instructor_firstname VARCHAR(20) NOT NULL,

instructor_lastname VARCHAR(20) NOT NULL,

instructor_email VARCHAR(40) UNIQUE NOT NULL,

Instructor_department_ id INTEGER

);

Authors & Contribution: Alyssa Phouangpanya 100%

Departments

CREATE TABLE Departments (

Department_id INTEGER PRIMARY KEY,

Deparment_prefix CHAR(4) UNIQUE NOT NULL,

Deparmtent_name VARCHAR(40) NOT NULL

);

Authors & Contribution: Aman Kalkat 100%

Courses

CREATE TABLE Courses (

Course_id INTEGER PRIMARY KEY,

Course_department_id INTEGER NOT NULL,

Course _code VARCHAR(10) UNIQUE NOT NULL,

Course_name VARCHAR(40) NOT NULL,

Course_credits INTEGER,

Course_description VARCHAR(255),

Course _catalog _year VARCHAR(10)

);

Authors & Contribution: Alyssa Phouangpanya 100%

Prerequisites

CREATE TABLE Prerequisites (

Prerequisite_ course_ id INTEGER,

Prerequisite _ prereq_ course_id INTEGER,

Prerequisite _requirement _ type VARCHAR(20)

);

Authors & Contribution: Aman Kalkat 100%

Enrollments

CREATE TABLE Enrollments (

Enrollment_ student_id INTEGER,

Enrollment_ section_id INTEGER,

Enrollment_grade VARCHAR(2),

Enrollment_ enrollment_date DATE

);

Authors & Contribution: Alyssa Phouangpanya 100%

Sections

CREATE TABLE Sections (

Section_id INTEGER PRIMARY KEY,

Section_ course_id INTEGER NOT NULL,

Section_ instructor _id INTEGER,

Section_ term VARCHAR(10) NOT NULL,

section_capacity INTEGER,

section_meeting_days VARCHAR(10),

section_start_time TIME,

section_end_time TIME

);

Authors & Contribution: Aman Kalkat 100%

Classrooms

CREATE TABLE Rooms (

Room_id INTEGER PRIMARY KEY,

Room_building VARCHAR(40) NOT NULL,

Room_number DECIMAL(3, 0) NOT NULL,

Room_capacity INTEGER NOT NULL

);

Authors & Contribution: Alyssa Phouangpanya 100%